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MILITARY SPECIFICATION

EXPLOSIVE, BOOSTER, R8150, FOR USE IN 120MM AMMUNITION

This specification is approved for use within the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1.0 SCOPE

1.1 Scope. - This specification establishes the requirements and quality acceptance provisions for the manufacture and acceptance of R8150 Explosive for use in 120mm ammunition.

1.2 Classification. - This specification covers R8150 explosive which is composed of wax coated RDX.

2.0 APPLICABLE DOCUMENTS

2.1 Government documents

* 2.1.1 Specifications, standards, and handbooks. - The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-R-398	RDX
MIL-W-20553	Wax, Desensitizing
MIL-A-48078	Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC-1376

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-650	Explosives: Sampling, Inspection and Testing
MIL-STD-1218	ACS Chemicals
MIL-STD-1235	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

* 2.1.2 Other Government documents, drawings and publications. - The following Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

* DRAWINGS (See 6.5)

* U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

7548644	Box, Packing for High Explosives, Assembly Details, Packing and Marking
7548645	Box, Fiberboard, Packing, Reusable, Collapsible for High Explosives

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. - The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is currently on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS PUBLICATION

ASTM	E300	Recommended Practice for Sampling Industrial Chemicals
ASTM	D2905	Statement on Number of Specimens Required to Determine the Average Quality of a Textile

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(Application for copies of ASTM Standards should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

* 2.3 Order of precedence. - In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3.0 REQUIREMENTS

3.1 Composition. - The composition shall be as follows when determined as specified in 4.5.1.

Ingredient	Specification	Composition (%)
RDX	MIL-R-398	95 ± 0.5
Wax (See 6.6)	MIL-W-20553	5 ± 0.5

3.2 Foreign matter. - The finished composition shall be free of visible impurities and foreign matter, when examined as specified in 4.5.2.

3.3 Moisture. - The moisture content of the composition shall not exceed 0.1 percent, when determined as specified in 4.5.3.

3.4 Acidity. - The acid content, as acetic acid, shall not exceed 0.02 percent, when determined as specified in 4.5.4.

3.5 Insoluble particles. - There shall be no insoluble particles retained on a United States (U.S.) Standard Number (No.) 40 sieve and not more than 5 insoluble particles retained on a U.S. Standard No. 60 sieve when determined as specified in 4.5.5.

3.6 Granulation. - The explosive shall comply with the following granulation requirements, when determined as specified in 4.5.6.

Through U.S. Sieve No.	Percent
10	100
50	5 max
100	2 max

3.7 Bulk density. - The minimum bulk density shall be 0.83 g/cc when determined as specified in 4.5.7.

3.8 Sensitivity. - The sensitivity value shall be determined for informational purposes only when the sample is tested in accordance with 4.5.8.

3.9 Preproduction inspection. - This specification contains provisions for the preproduction inspection. Requirements for the submission of preproduction samples by the contractor shall be as specified in the contract (see 6.3).

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* 3.10 First article. - When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3).

* 3.11 First article inspection. - This specification makes provision for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

* 3.12 Workmanship. - All parts shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, surface defects, flash, air bubbles, blisters and dirt. The cleaning method shall not be injurious to the parts, nor shall the parts be contaminated by the cleaning agent. (See 4.5.9)

4.0 QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. - The inspection requirements specified herein are classified as follows:

* a. First article inspection (see 4.3)
* b. Quality conformance inspection (see 4.4)

4.3 Preproduction sample inspection. - The preproduction sample tests shall provide the necessary information to determine that the explosive meets the chemical, physical and performance parameters defined herein.

4.3.1 Submission. - The contractor shall submit a first article sample, as designated by the Contracting Officer, for evaluation in accordance with provisions of 4.3.2. The first article shall consist of 4.5 kg (10 pounds) of the composition obtained from a production batch which has been produced by the contractor using the same production processes, procedures and equipment as will be used in fulfilling the contract. All materials shall be obtained from the same sources of supply as will be used in regular production.

4.3.2 Inspection to be performed. - The sample will be subjected by the Government to any or all of the examinations or tests specified in Table I.

4.3.3 Rejection. - See MIL-A-48078.

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4.3.4 Process control. - Submission of process controls to be utilized by the manufacturer to assure a quality product shall be made to the technical agency for review on all critical defects not subject to direct, discrete measurement and all major defects specifying sample plans requiring fewer samples than the routine inspection levels of MIL-STD-105 or MIL-STD-1235.

4.4 Quality conformance inspection

4.4.1 Lot formation. - A lot shall consist of one or more batches of explosive, produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each batch shall consist of that quantity of explosive that has been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous. In addition, each lot shall contain RDX from one manufacturer only.

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CLASSIFICATION OF CHARACTERISTICS AND TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER / NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Composition			3.1	4.5.1
	Foreign Matter			3.2	4.5.2
	Moisture			3.3	4.5.3
	Acidity			3.4	4.5.4
	Isoluble Particles			3.5	4.5.5
	Granulation			3.6	4.5.6
	Bulk Density			3.7	4.5.7
	Sensitivity (info only)			3.8	4.5.8
	Workmanhip			3.12	4.5.9

NOTES:

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Wooden or Fiberboard Box	1 OF 1	7548644/7548645
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Foreign matter		100%		Visual
102	Liner pierced or torn		100%		Visual
103	Liner improperly closed		100%		Visual
Minor					
201	Type of liner incorrect		100%		Visual

NOTES:

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 7548644
4.4.2.2	Sealed Wooden Box				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Box damaged		100%		Visual
102	Lot number incorrect or illegible		100%		Visual
103	Board broken or split		100%		Visual
104	Strapping missing, broken or loose		100%		Visual/Manual
105	Top improperly assembled		100%		Visual/Manual
Minor					
201	Nail protruding		100%		Visual
202	Marking incorrect or illegible		100%		Visual
203	Strapping improperly assembled		100%		Visual/Manual

NOTES:

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Sealed Fiberboard Box				7548645
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Assembly torn or pierced		100%		Visual
102	Lot number incorrect or illegible		100%		Visual
103	Banding strips broken, missing or loose		100%		Visual/Manual
Minor					
201	Marking incorrect or illegible		100%		Visual
202	Stitches missing or loose (when required)		100%		Visual/Manual
203	Banding strips improperly assembled		100%		Visual/Manual

NOTES:

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4.4.2 Examination. - Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II (see MIL-A-48078).

4.4.3 Testing

PRECAUTION WARNING

This specification covers sampling and testing of toxic or hazardous materials. Accordingly, it is emphasized that all applicable safety rules, regulations, and procedures must be followed in handling and processing these materials.

4.4.3.1 Sampling. - Approximately 300 grams of the composition shall be selected from each batch to be sampled using ASTM Procedures E300 for solids. Samples shall be selected for inspection in accordance with MIL-STD-1235, CSP-1 Plan, Sample Frequency Code Letter B, AQL 6.5%. If any sample fails to meet any test requirement, the batch represented by the sample shall be rejected. All batches produced between the last batch that was tested and accepted and the batch which failed shall be tested in accordance with the applicable methods given in paragraph 4.5. If any of these batches fail to meet any of the test requirements, that batch shall also be rejected. In addition, after any failure of a batch, the contractor will return to 100% inspection until "i" successive batches are accepted as required by MIL-STD-1235. The defect classification shall be as given in Table II.

TABLE II. Classification of Defects.

Inspection/Requirement	Defect Classification
Composition	Major
Foreign Matter	Major
Moisture	Major
Acidity	Major
Insoluble Particles	Major
Granulation	Major
Bulk Density	Major
Sensitivity	Information Only
Workmanship	Major

4.4.4 Inspection equipment. - Commercially available inspection equipment should be used in the performance of all tests and examinations specified in 4.4 and 4.5. The contractor shall have the required equipment available, follow proper calibration methods, and perform the testing in accordance with accepted operating procedures. Government approval of all inspection equipment is required prior to its use for acceptance purposes (see 6.3).

4.5 Test methods and procedures (see 6.4). - The following tests shall be performed using prescribed analytical procedures for replicate determinations given in standard analytical text books. ASTM method D2905 may also be used to

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determine the number of replicates required. Unless otherwise specified herein, all chemicals and reagents used in test methods shall be ACS grade in accordance with MIL-STD-1218 or reagent grade.

4.5.1 Determination of composition.

4.5.1.1 RDX content (see 6.4). - A medium porosity glass filtering crucible shall be cleaned with hot benzene followed by a final rinse of acetone. The crucible shall be backfiltered with acetone and be free flowing. The crucible shall be dried, weighed on an analytical balance and held in the oven at 60-70°C until used. A weighed portion of sample shall be placed in a 150 ml beaker with a magnetic stirring bar and 60 ml of aliphatic naphtha solvent shall be added. The beaker shall be covered with a watch glass and placed on a magnetic stirrer hot plate (or equivalent) maintaining the temperature of the solution between 80 and 90°C and stirring for 30 minutes. The hot plate shall be checked for proper temperature with a thermometer in a 400 ml beaker containing 100 ml of solvent. The top of the beaker shall be covered with aluminum foil and the solvent temperature shall be maintained within the specified range. During the filtration operation the previously prepared crucible and contents shall be kept at 60-70°C in order to prevent wax from separating from the solution and clogging the crucible. This may be accomplished by use of a heat gun or other commercially available equipment. The top of the solvent wax shall be decanted through the crucible and the RDX remaining in the beaker (and on the stirring bar) shall be washed with warm aliphatic naphtha solvent and transferred quantitatively to the crucible using a wash bottle. No more than 40 ml of aliphatic naphtha solvent shall be used to effect the transfer. If necessary, rinse beaker with hexane to remove RDX crystals that cling to sides of the beaker. During filtration, the crucible shall not be aspirated dry until the final wash in order to prevent the wax from freezing-out on the crucible. The crucibles shall be rinsed twice with 10 ml portions of hexane. The RDX shall be stirred with hexane before applying a vacuum. The crucible and contents shall be dried for 15 minutes in an oven (90-105°C) or until constant weight is obtained. The percentage of RDX shall be calculated as follows:

$$\text{Percent RDX} = \frac{100A}{W-(MW)}$$

Where:

A = increase in weight of filtering crucible, gms.
W = weigth of sample, gms.
M = percent moisture in material, expressed as a decimal (see 4.5.2).

4.5.1.2 Wax (see 6.4.1). - The percentage of desensitizer shall be calculated on a moisture free basis by subtracting from 100% the percentage of RDX obtained as specified in 4.5.3.1.

4.5.2 Determination of foreign matter. - Spread the sample taken in 4.4.3.1 on a white flat surface. Visually examine the material for foreign matter using daylight illumination to determine compliance with 3.2.

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4.5.3 Determination of moisture content. - See 6.4.1.1.

4.5.3.1 Special solvent. - The special solvent shall consist of 80 parts methanol and 20 parts naphtha by volume thoroughly mixed. If necessary, the solvents shall be dried by distillation.

4.5.3.2 Procedures (Karl Fisher Method). - Determine the moisture content in accordance with Method 101.4 of MIL-STD-650.

4.5.4 Acidity. - A 6 gram specimen shall be weighed accurately to the nearest 0.001g and transferred to a 400 ml beaker. A 25 ml portion of aliphatic naphtha shall be added and the beaker and contents covered with a watch glass and heated on a steam bath until the coating on the RDX crystals has completely dissolved. The aliphatic naphtha solution shall be decanted carefully into a clean small beaker, covered with a watch glass, and kept warm (above 40°C). Seventy-five ml of acetone shall be added to the RDX crystals remaining and the 400 ml beaker covered and heated on a steam bath until all of the crystals are in solution. The aliphatic naphtha solution shall be poured carefully back into the acetone solution and the small beaker washed thoroughly with about 5 ml of aliphatic naphtha, which shall also be added to the acetone solution. The beaker shall be heated gently while stirring to thoroughly mix the solution. Add 200 ml of cool distilled water to the solution while continuing to stir. Remove the solution from the steam bath and titrate with 0.05N sodium hydroxide solution using either methyl red or bromothymol blue as an indicator. A blank determination shall be made using the same amounts of reagents. The percentage of acidity as acetic acid shall be calculated as follows:

$$\text{Percent acidity} = \frac{6.005\ (V-v)N}{W-(MW)}$$

where:

V = milliliters sodium hydroxide used in titrating specimen
v = milliliters sodium hydroxide used in blank
N = normality of sodium hydroxide solution
W = weight of sample in grams
M = percent moisture in materials expressed as a decimal (see 4.5.3)

4.5.5 Insoluble particles. - (See 6.4.1.3) Determine the insoluble particles in accordance with Method 106.1 of MIL-STD-650.

4.5.6 Granulation. - Determine the granulation in accordance with Method 204.1 of MIL-STD-650.

4.5.7 Bulk density. - Determine the bulk density in accordance with Method 201.1.1 of MIL-STD-650.

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4.5.8 Sensitivity

SAFETY PRECAUTIONS

Observe normal safety precautions. Pay particular attention to the following:

1. Operate the exhaust fan while tests are being performed.
2. Do not operate the machine without a safety shield.
3. Secure the weight with a safety chain while cleaning the machine.
4. Remove waste explosives from the room after completion of each test.
5. Compliance with local safety rules is required.

4.5.8.1 Apparatus. - Impact machine (Naval Ordnance Drawing LD-70518), Sandpaper (No. 05, approximately 1 inch squares), 35 mg loading spoon wood, or hard rubber spatula.

4.5.8.2 Procedure. - Pass a dry sample of the explosive through a sample splitter until a representative portion of approximately 10g is obtained. Proper safety should be exercised in determining the impact sensitivity. Position approximately 25 sandpaper squares with the abrasive side up on a flat surface. Fill the loading spoon with dry composition and smooth off the excess by drawing a spatula across the flat surface of the spoon. Pour the remaining portion upon the center of one of the sandpaper squares. Repeat this procedure until all sandpaper blocks are loaded. Turn the exhaust fan switch to the "ON" position to remove dust and fumes from the area of the impact machine. Place the impact weight on the weight retainer. NOTE: Ascertain the anvil and plunger are clean, free of scratches and cracks. Place one of the prepared samples on the center of the anvil. Insert the plunger through the guidehole above the anvil and allow the plunger to rest gently on the sample. Turn the electromagnet switch to the "ON" position. Lift the weight vertically until it is held in place by the electromagnet. Adjust the height of the weight by turning the gear handle to a height, that when dropped, will explode or burn the sample. Remove the weight retainer from the path of the weight. Facing away from the impact machine, turn the electromagnet switch to the "OFF" position, allowing the weight to fall and strike the top of the plunger. Lift the weight, replace the weight retainer and allow the weight to rest upon the retainer. Remove the plunger and determine if an explosion has occurred. Clean all unexploded material from the anvil, plunger, and base plate with a brush or cloth. NOTE: An explosion may be recognized by a sharp report, smoke in the area of the plunger or disintegration of the sample. Use of instrumented microphone with detectors is recommended. Once again place the impact weight on the weight retainer, place another prepared sample on the anvil and repeat the procedure, raising the electromagnet 5cm after each non-explosion and lowering the electromagnet 5cm after each explosion. The first non-explosion after an explosion is considered as the starting point of the 20 tests. Record this height in cm. Opposite the height recorded for the starting point, place a minus (-) sign designating a non-explosion. Raise the electromagnet 5cm and place the weight retainer in place and again repeat the procedure. If an explosion occurs, place a plus (+) sign opposite the height tested. If a non-explosion occurs, place a plus (+) sign opposite the height

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tested. If a non-explosion occurs, place a minus (-) sign. Raise or lower the electromagnet as required and repeat the procedure until 20 tests have been completed. Record each test result.

Calculate the impact sensitivity as follows:

$$\text{Impact sensitivity, cm} = C - \frac{5\,(D-50)}{DE}$$

Where C = The lowest height in cm at which more than 50% explosions occurred.
D = Percentage of explosions at height C
E = Percentage of explosions at height 5 cm less than C
5 = Difference in height in cm of each test

Example:

45 cm = 80% explosions
40 cm = 30% explosions
C = 45 cm
D = 80% explosions
E = 30% explosions
5 = 45 cm - 40 cm (difference)

Calculate the impact sensitivity as follows:

$$\text{Impact sensitivity, cm} = C - \frac{5\,(80 - 50)}{80-30}$$

$$= 45 - \frac{5 \times 30}{50}$$

$$= 45 - 3$$

Impact sensitivity = 42 cm

4.5.8.3 Interferences

1. A machine loosely assembled or not in proper alignment may produce incorrect values.

2. A rough surface or cracks on the anvil or plunger may produce lower sensitivity values.

3. An insufficient or unevenly distributed sample may produce incorrect values.

4. Samples containing glass, metal or other gritty matter foreign to the product may produce low sensitivity values. (A low sensitivity value indicates more sensitive materials.)

5. Samples containing moisture, oil, grease and/or soft plastic may produce high sensitivity values.

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4.5.9 Workmanship. - Visually examine the R8150 explosive to verify compliance with the requirements of 3.10.

5.0 PACKAGING

5.1 Packing

5.1.1 Level A. - Unless otherwise specified, the Composition shall be packed in accordance with drawing 7548644.

* 5.1.2 Level B. - Unless otherwise specified, the composition shall be packed in accordance with drawing 7548644.

* 5.1.3 Level C. - (Level C to be used for CONUS shipment only.) Unless otherwise specified, the composition shall be packed in accordance with drawing 7548645.

5.2 Marking

5.2.1 Level A. - Marking shall be in accordance with drawing 7548644.

* 5.2.2 Level B. - Marking shall be in accordance with drawing 7548644.

* 5.2.3 Level C. - Marking shall be in accordance with drawing 7548645. The top and one side of the container shall also be marked "CONUS SHIPMENT ONLY" in addition to the other required marking.

* 5.3 Shipping. - When composition from more than one lot is shipped at the same time, each lot shall be kept separate and the division between the lots clearly indicated to prevent mixing of the lots in transit.

6.0 NOTES

6.1 Intended use. - The explosive composition covered by this specification is intended to be used as an explosive for 120mm ammunition.

6.2 Ordering data. - See MIL-A-48078 (PA).

* 6.3 Submission of inspection equipment designs for approval. (see MIL-A-48078). - Submit equipment designs, as required to Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAT-I-(D), Dover, New Jersey 07801-5001.

6.4 Equivalent test methods. - The test methods given in this specification are the official methods to be used. The contractor may request using other methods providing that the proposed method is equivalent (accuracy and precision) to the method given in this specification. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to: Commander, ARDEC, ATTN: AMSMC-QAR-R, Dover, NJ 07801-5001. This

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description should include, but not be limited to, the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy, and precision and drawings of any special equipment required.

6.4.1 Approved equivalent test methods. - The following methods previously submitted by Holston Defense Corp are considered equivalent test methods.

6.4.1.1 Moisture content. - ASM No. I-7.

6.4.1.2 Sensitivity. - ASM No. P-57 W/P-54.

6.4.1.3 Insoluble particles. - ASM No. C-4.

6.5 Drawings. - Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal U.S. Army Armament Research and Development Command drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.6 Approved material. - BARECO No. ES409.

* 6.7 Definition of Packaging and Packing Levels

* 6.7.1 Level A. - Full military protection for unrestricted (CONUS) shipment and storage.

* 6.7.2 Level B. - Limited military protection for limited (CONUS) shipment and storage.

* 6.7.3 Level C - For interplant (CONUS) shipment and very limited storage.

* 6.7.4 For further detailed information concerning packaging and packing levels, consult AR-700-165."

* 6.8 Keywords. -
Ammunition
Cartridge
Cannon
Explosive, Booster, R8150
Tank

* 6.9 The margins of this standard are marked with asterisks to indicate where significant changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations.

Custodian:
Army-AR

Preparing Activity:
Army-AR

Project Number: 1376-A310

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